Consumer Valuation of Battery Health Information in Used Electric Vehicle Markets: A Discrete Choice Experiment

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ABSTRACT

This discrete choice experiment (DCE) aimed at understanding consumer preferences and willingness to pay for battery health information in the used battery electric vehicle (BEV) market. The potential survey design includes questions about current vehicle information, future vehicle preferences, a discrete choice experiment (DCE), EV knowledge, and demographics. The DCE systematically varies both standard vehicle attributes (e.g., price, mileage) and battery-specific features (e.g., state-of-health, refurbishment history) to quantify the economic value consumers place on battery-related information. Findings from this research project will improve the understanding of second-life battery markets, and guide future research in the evolving used EV landscape.

1. Introduction

used vehicles make up in the U.S. (Bureau of Transportation Statistics 2023 New and used passenger car and light truck sales and leases (available at: www. bts.gov/content/new-and-used-passenger-car-sales-andleases-thousands-vehicles))

Electric vehicles (EVs) are gaining momentum in the automotive market. While much of the focus of industry professionals and academics has been on promoting new EVs, the used EV market has received comparatively little attention. Meanwhile, the used vehicle market in the U.S. has reached record highs in recent years reaching 70% of all vehicle purchases and is projected to grow steadily over the coming decade (Market Research Future, 2025). As more EVs reach the end of their initial ownership, they are increasingly entering the used market. According to February 2025 sales data (Cox Automotive Inc., 2025), used EV sales rose by 34.2% year-over-year, while the average days' supply fell by 21.5%, signaling strong and accelerating demand.

The growth of the used EV market is crucial from environmental, economic, and market development perspectives. It contributes to a circular economy by extending the useful life of vehicles and batteries, thereby reducing lifecycle emissions (Guzek et al., 2024). Used EVs also lower the financial barriers for middle- and lower-income households to enter the EV market (Hagman et al., 2016), although the affordable models remain relatively limited in availability (Sheykhfard et al., 2025). Furthermore, a healthy resale market boosts value retention of new EVs, enhancing consumer confidence in their long-term investment (Brückmann et al., 2021; Lim et al., 2015).

To support future research and strategy development on the acceptance of used EVs requires a deeper understanding of consumer preferences and concerns. Despite rising supply, consumers still perceive used EVs as scarce in the marketplace (Krishna, 2021). Survey data (Fernandes, 2023) also reflect lingering consumer hesitation: 52% of U.S. adults reported being unlikely to consider buying a used EV, compared to only 28% who were open to the idea. In addition to challenges common to both new and used EV, such as range anxiety, limited charging infrastructure, and high purchase costs (Sheykhfard et al., 2025), used EVs come with added concerns. Key issues include uncertainty about the vehicle history and condition (cited by 41% of respondents in Fernades' study), as well as perceptions of

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lower reliability compared to new EVs (23%). Based on Sheykhfard et al. (2025), more than half (55.1%) of current used EV owners reported having pre-purchase concerns about battery longevity, primarily due to the perceived risk of frequent and costly battery replacements. However, while nearly two thirds (65.5%) of respondents reported some degree of battery decline over time, only 5.1% experienced a substantial loss in battery performance. Moreover, 39.2% and 26.3% of them indicated they are likely or very likely, respectively, to buy a used BEV in the future. These findings suggest that the pre-purchase concerns appear to be overstated in practice, and most used EV owners had a positive ownership experience.

These studies highlight that the used EV market faces a significant information asymmetry problem centered on battery: the most critical and expensive component affecting a used EV's value, in terms of its unobserved maintenance history, current "State of Health" (SOH), and remaining useful life (RUL). Unlike conventional vehicles, where age and mileage serve as reasonable proxies for engine condition and overall wear (Prieto et al., 2015), the battery degradation path and RUL of the battery can vary substantially even among EVs of the same model year and mileage, making the vehicle's true condition far more difficult to assess and increasing the level of risk in the purchase for consumers. This information gap creates economic inefficiencies. Buyers may undervalue well-maintained EVs or unknowingly overpay for poorly maintained ones. Sellers with well-maintained batteries may struggle to recoup the true value of their vehicles at resale. More broadly, market uncertainty may dampen new EV sales, as buyers hesitate to purchase due to "resale anxiety". Understanding the key determinants of used EV purchase decisions, particularly for battery electric vehicles (BEVs), and how consumers evaluate battery health information is critical for policymakers, industry stakeholders, and researchers. This study would be the first attempt to quantify the economic value consumers place on battery health information (i.e., WTP) in the used BEV market. At the core of the study is a discrete choice experiment (DCE), which presents respondents with repeated choices between hypothetical used BEVs. Drawing on insights from the existing literature, the research team identifies key factors influencing used BEV adoption. The DCE systematically varies both conventional vehicle attributes (e.g., mileage, purchase price) and battery-specific attributes (e.g., range, refurbishment history, SOH) to determine which characteristics most influence consumer preferences and to estimate their WTP for specific aspects of battery performance. Ultimately, this study supports future advancements in BEV technology, the effective communication of critical battery information at the point of sale, and a better understanding of consumer priorities, demographic differences, and decision-making factors in the used BEV market.

The remainder of the paper is structured as follows. Section 2 provides a review of literature on household vehicle composition and usage (among BEV users), as well as factors influencing BEV adoption. Section 3 provides an overview of the survey data, and presents both descriptive statistics and comparative analyses. Section 4 outlines the statistical method employed in the analysis. Section 5 presents the key findings, discusses their implications, outlines study limitations, and suggests directions for future research. Finally, Section 6 concludes the paper.

2. Literature Review

This section reviews key literature informing survey design and analyses of this study. The first subsection examines the main drivers and barriers to the adoption of used BEVs, while the second subsection focuses on EV battery technology and how it shapes consumer perceptions and decision-making.

2.1. Used Electric Vehicle Adoption

Gore et al. (2024) provided a comprehensive review on consumer perspectives on EVs and the factors influencing EV adoption, identifying a broad range of determinants including attitudinal, demographic, societal, environmental, technological, and economic variables (Iogansen et al., 2023; Naseri et al., 2024; Sonar et al., 2023). However, most existing studies focused on new EV markets, with comparatively limited attention given to the used EV markets, which is rapidly expanding and playing an increasingly important role in the broader diffusion of electric mobility.

Though some preferences may carry over between new and used vehicle buyers—for instance, longer driving range consistently correlates with stronger EV preference—emerging evidence suggests that these two consumer groups differ in meaningful ways. For instance, used EV buyers are more sensitive to the availability of public charging infrastructure than new EV buyers, likely due to lower rates of home charging access (Zou et al., 2020). Moreover, purchasing behavior for used vehicles is often more risk-averse and budget-conscious, reflecting different decision-making processes than those found in the new car market. This indicates that used EV adoption is not simply a scaled-down version of new EV adoption and warrants independent investigation. Several studies have begun to profile the typical used EV buyer. Canepa et al. (2019) found that used car buyers are commonly cost-conscious. Compared to the general population, used EV buyers are more likely to be younger, male, White, and better educated. However, compared to those new-EV buyers, used EV buyers tend to have lower household incomes and be renters, with lower access to private garages with EV chargers (Khaloei et al., 2020; Sheykhfard et al., 2025). These characteristics differ from the traditional early adopter profile seen in the new EV market (typically wealthier, homeowners with garages), suggesting that used EVs may serve as a more accessible entry point into EV ownership for broader population segments.

Although range is a common concern for EV users, Sheykhfard et al. (2025) found that the majority of used EV owners drive well within the range capabilities of most existing EV models: 59.4% reported driving fewer than 100 miles per day and 29.4% drove less than 50 miles per day. This implies that range anxiety may be less of a barrier for used EV buyers, who tend to have shorter commutes or use their vehicles in more limited travel contexts. Therefore, range may serve as a less decisive factor in used EV adoption compared to other attributes such as battery condition, charging accessibility, and price.

Electric Vehicle Battery Technology and Consumer Perception The electric vehicle batteries (EVBs) present new risks and opportunities for consumers and the automotive industry. Despite progress in battery technology, consumer concerns about battery health, replacement cost, and resale value persist, especially in the used EV market. A major concern is battery cost, which can represent up to 30% of an EV's total price (Boudway, 2020). Because of this, battery longevity and reliability play a crucial role in determining the economic viability of EV ownership and attractiveness of used EV.

2.2. Battery Degradation and State-of-Health

In the adoption literature, battery-related considerations for BEVs have largely centered on driving range (Yuan et al., 2018). Similarly, the common hesitation among prospective used EV buyers stems from battery degradation, SOH uncertainty and the unreliability of older technology (Pedrosa and Nobre, 2018). Battery degradation is typically manifested by a decrease in energy capacity, driving range, and increased charging time over time (Attia et al., 2022). The rate of degradation depends on multiple factors, including charging patterns (e.g., fast versus slow charging, charging frequency), driving patterns, depth of discharge cycles, and operation conditions (e.g., climate and temperature exposure) (Bashash et al., 2011; Neubauer et al., 2012). Yang et al. (2018) suggests that the BEV batteries can last five years to thirteen years depending on the climate. Studies suggest that battery aging trajectories are often linear or sublinear, meaning that the degradation either occurs at a constant rate or a reduced rate over time, likely due to the stabilization of battery chemistry after initial use (Attia et al., 2022, 2020). Batteries are generally considered to reach their End-of-Life (EoL) for vehicle use when their SOH drops to approximately 70–80% of the original capacity (Canals Casals et al., 2022, 2019). Battery degradation has been shown to reduce consumers' perceived future value of the vehicle and heighten resale anxiety.

2.3. Battery Replacement, Cost, and Warranty Considerations

While the existing literature has considered the longevity of the batteries when calculating the overall cost of EVs (Hagman et al., 2016; Letmathe and Suares, 2017), there is a lack of studies that directly examined the cost of EVBs. Most modern EVBs are designed to last the lifetime of the vehicle under normal use. Dnistran (2024) studied almost 5,000 EVs and revealed that the average annual degradation rate to be just 1.8%, making the battery lifespan often outlasts the vehicle's useful life. However, Sheykhfard et al. (2025) report that although battery performance loss to the point that they need to be replaced are rare but can be expensive. Drawing from real-world experience compiled from a few online sources, the replacement costs can range between \$5,000 and \$20,000 or more depending on the vehicle model and labor (Kothari, 2024; Witt, 2024).

To improve consumer confidence, most EV manufacturers offer warranties covering 8 years or 100,000 miles —- whichever comes first (Clarke, 2024). This suggests that consumer purchase intentions can be closely linked to how long they expect to keep the vehicle and whether the battery's SOH will fall below the threshold (70% 75%) within the ownership window. Hitting this threshold can trigger a free battery replacement, which adds economic value and reassurance. This is supported by prior literature suggesting that the cost of BEVs varies widely by how long someone is expected to own the vehicle (Hagman et al., 2016). Letmathe and Suares (2017) also specifically consider reselling the battery separately from the resale cost of the vehicle when calculating the total cost of ownership .

Second-Life and Refurbished Batteries As the number of used EVBs increases, opportunities for reuse and recycling become critical components of a circular battery economy (Skeete et al., 2020; Tankou et al., 2023). Many circular business models emphasize two main second-life pathways for EVBs: one is to exploit deteriorated batteries for less demanding applications, such as stationary energy storage for homes or grids (Christensen et al., 2021), and the other one is to refurbish those batteries and reintegrate into the EV market. This review focuses on the latter — refurbishing batteries for continued use in EVs.

Refurbishing batteries at the cell or module level can offer cost savings and support the economic viability of BEVs. For example, Jiao and Evans (2016) suggest that repurposing EoL EVBs could reduce the cost differential between BEVs and conventional vehicles. Similarly, Shaikh et al. (2023) conducted a simulation showing that refurbished EVBs are more affordable than new ones. Moreover, the refurbishment costs are lowest when batteries are sold locally, compared to regional and national markets. These findings indicate clear advantages for both automotive manufactures and customers. However, existing studies on second-life EVBs have largely focused on technical and economic feasibility, often outside of the US contexts where EV penetration is higher and EoL battery volumes are projected to increase significantly. In contrast, relatively little attention has been paid to consumer preferences and WTP for refurbished EVBs, which are essential to the success of refurbishment-based business models.

One of the few relevant studies, conducted by Pedrosa and Nobre (2018) in Portugal found that all interviewees expressed positive attitudes toward refurbished EVs, particularly those fitted with replacement batteries backed by dealership warranties. Participants viewed these vehicles as more reliable, with the potential for extended driving range and greater peace of mind, and even indicated a higher WTP compared to used EVs retaining their original batteries. The main limitation of this study is that it is a qualitative interview based on a very small sample and may not be generalizable to the US market. Moreover, while the study suggests a consumer preference for BEVs with brand-new batteries than original ones, to the best of the authors' knowledge, no studies have directly examined consumer attitudes towards more nuanced refurbishment options, such as partial battery replacements where only some cells are replaced to address performance issues or extend battery life while the rest cells remain original, or full pack replacements where the entire battery is swapped with a new pack after the original battery's performance declines below a certain threshold. Understanding consumer acceptance of these varying levels of refurbishment is essential for informing viable second-life battery strategies in the used EV market.

2.4. Willness to Pay for Battery Information

To be continued.

3. Survey Design, Data Collection, and Sample Profile

3.1. Used EV Survey

The survey was designed to understand consumer preferences and willingness to pay for vehicle attributes and battery health information in the used BEV market. The survey targets on US consumers who is 18 years old and has the intention to purchase a used car or SUV within the next two years.

The core component of the survey is the BEV discrete choice experiments where respondents are asked to consider a scenario in which they choose a used BEV from a set of options with varying attributes. It is designed to capture trade-offs that consumers make when evaluating used BEVs, particularly how they value battery health information relative to other vehicle attributes. Each respondent completes six repeated choice tasks. Before presenting the choice tasks, we introduce the specific attributes of the BEVs that respondents will evaluate in the upcoming choice experiments (see Table 1). Based on prior studies, age and mileage are the main factors influencing the value of a used car (Prieto et al., 2015). For all choice questions, we ask respondents to assume the used vehicle was built in 2022 and is currently three years old. The attributes with varying values include vehicle mileage, purchase price, battery refurbishment history, electric range, and battery state of health. All attribute descriptions are provided to ensure respondents clearly understand the meaning of each before making their selections.

To examine the effects of information treatment, specifically, how the presentation of certain information may influence consumer preferences, participants are randomly divided into two groups. One group receives only information about vehicle driving range and battery health, while the other group is shown additional details about battery maintenance and warranty coverage. This design allows us to assess whether specific types of information impact respondents' choices in the BEV selection tasks. Figure 1 provides an example of the choice experiment, where respondents are asked to choose between three used BEVs with varying attributes and a "None of the above" option.

Table 1
BEV choice experiment attributes and levels.

Attribute	Definition	Levels			
		Low-budget (≤$20K) Car	**High-budget** (>$20K) Car	**Low-budget** (≤$20K) SUV	**High-budget** (>$20K) SUV
Mileage	The total number of miles a vehicle has traveled since manufacture.	15,000–60,000 miles, in increments of 500 miles.			
Battery refurbishment history	The repair or replacement work that has been performed on the vehicle's main battery pack.	*(1) Original:* The battery remains in its factory-original condition with no repairs or replacements. *(2) Some battery cells replaced:* A portion of the battery cells has been replaced to address performance issues or extend battery life; the remaining cells are original. *(3) Entire battery pack replaced:* The full battery pack has been replaced with a new, refurbished, or used pack, typically after battery performance declined below a specified threshold.			
Electric range at delivery (Year 0)	The maximum distance the vehicle can travel on a full battery charge under typical driving conditions, as rated at the time of manufacture.	50–150 miles (50-mile increments)	100–250 miles (50-mile increments)	150–250 miles (50-mile increments)	200–350 miles (50-mile increments)
Battery annual degradation rate and state of health (SOH)	The average annual percentage loss in electric range and battery SOH. The remaining range and SOH at Year 3 (current) and Year 8 (five years from purchase) are derived from this rate.	1%–8%, in increments of 1%.			
Purchase price	The total cost of the vehicle in dollars, including down payment, monthly payments, taxes, and fees.	$10,000–$20,000 ($2,000 increments)	$20,000–$40,000 ($5,000 increments)	$15,000–$25,000 ($2,000 increments)	$25,000–$45,000 ($5,000 increments)

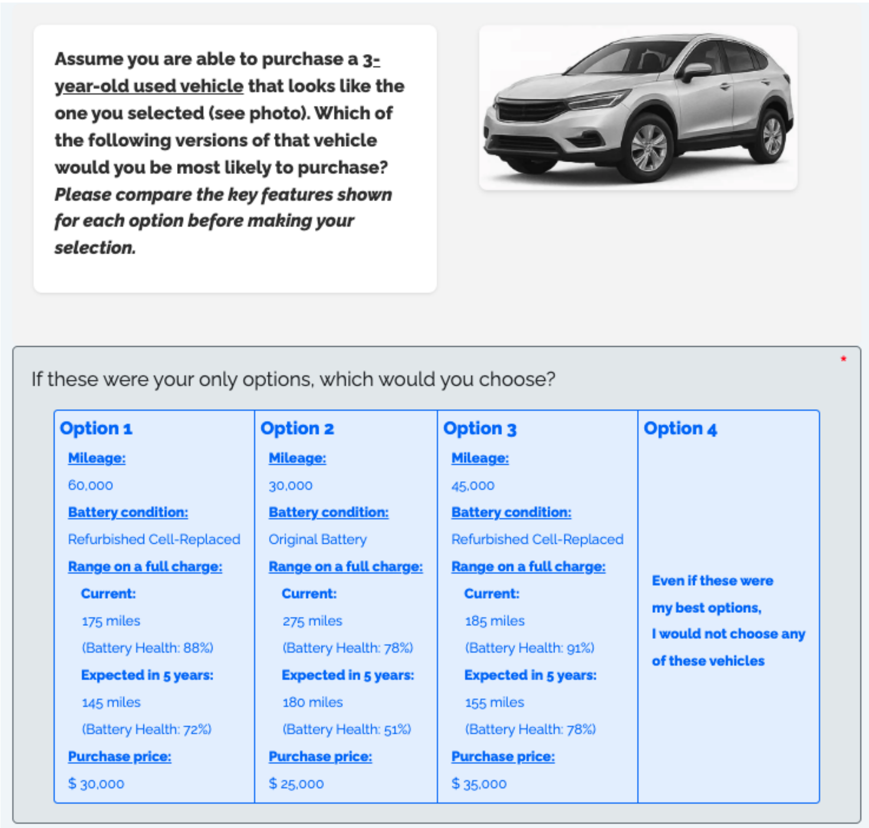

Figure 1: Example of a choice experiment.

In addition to DCEs, the survey gather information on a range of other topics. Before the DCEs, The survey collects information on household's current vehicle fleet and detailed vehicle attributes for their primary vehicle if they report owning or leasing at least one (fuel type, fuel efficiency, range,new/used, way of obtain, purchase/lease price,monthly payment,refuel/recharge frequency), and their future vehicle purchasing plans and preferences (anticipated budget,payment method, andthe likelihood of purchasing a new or used plug-in hybrid or BEV). After the DCEs, the survey assesses respondents' knowledge, attitudes, and perceptions related to EVs, battery technologies, and related topics. It begins with a series of factual questions that evaluate understanding of which vehicles can run on gasoline or be plugged in, whether respondents can name a BEV, and their knowledge of the current U.S. federal tax credit for new EV purchases. Respondents then rate their agreement with a series of attitudinal statements in 5-point Likert scale covering topics such as social norms, environmental beliefs, cost perceptions, battery concerns, and personality traits like risk-taking and price sensitivity. The final section of the survey collects demographic and additional attitudinal information. It includes standard demographic questions such as year of birth, gender, race/ethnicity, household size, income, education level, employment status, and housing type. These variables are crucial for understanding how socio-demographic factors influence preferences and attitudes toward EVs. Additional questions explore home ownership and average electricity bills, factors that may shape EV adoption decisions.

For a comprehensive description of the survey design, see the Data Collection Instrument (Gore et al., 2026).[7]

3.2. Data Collection and Sample Profile

The survey was administered online through two widely-used online research platforms. The data collection took place between December 2025 and April 2026. After initial data cleaning, we ended with 2,113 participants across the US. The geographic distribution of respondents is shown in Figure 2. Each point represents a ZIP code, with color and size scaled to respondent count. The median survey time is around 14 minutes. The survey participants were compensated through the research platforms.

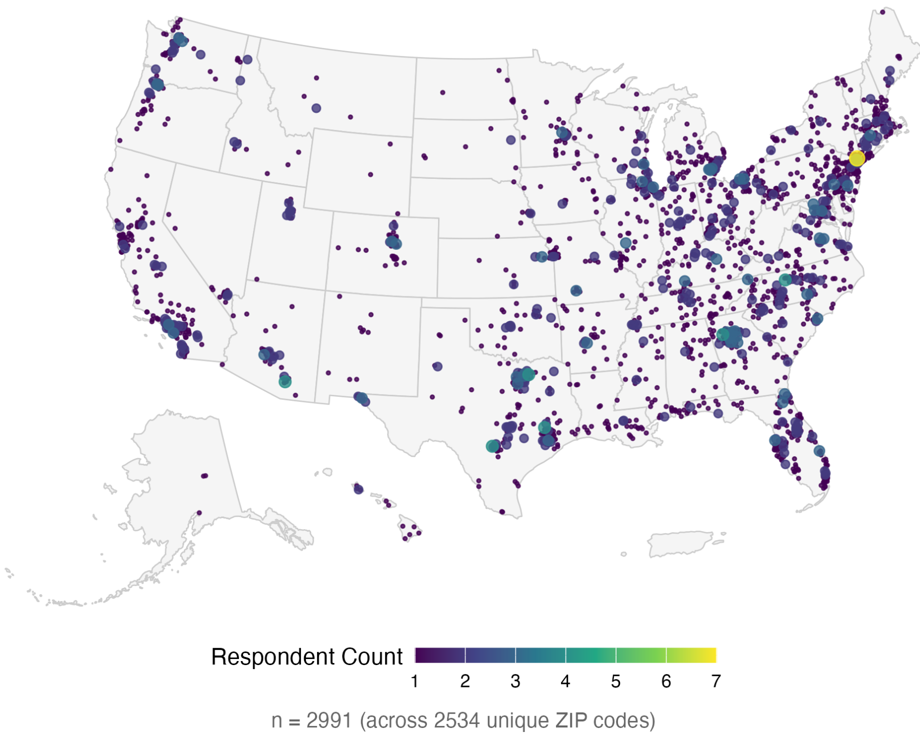

Figure 2: Geographic distribution of survey respondents across the continental United States.

[7]The Data Collection Instrument was published prior to the start of data collection. Certain aspects of the survey design were revised following piloting. Where discrepancies exist between the instrument and the present paper, the description in this paper reflects the final implemented version.

We further apply multiple data quality screens to identify and remove low-quality responses, including failure of an embedded attention check, inconsistency between reported information in different sections of the survey, uninformative filler text and gibberish in open-ended responses, and so forth. In the end, the final sample size for analysis is 1,500 respondents.

4. Method

4.1. Mixed Logit Model (MXL)

The Mixed Logit (MXL) model relaxes the restrictive Independence of Irrelevant Alternatives (IIA) assumption of the standard MNL by allowing preference parameters to vary continuously across individuals (**mcfaddenMixedMNLModels200** **trainDiscreteChoiceMethods2009**). We estimate the model directly in *WTP space* (**trainWTPSpacePreference2005**). Partitioning the attribute vector into price p_{nti} and non-price attributes $\mathbf{x}_{nti}$, the WTP-space utility for respondent n choosing alternative i in choice scenario t is:

$$U_{nti} = -\alpha_n \left(p_{nti} - \mathbf{w}_n' \mathbf{x}_{nti} \right) + \varepsilon_{nti} \tag{1}$$

where $\alpha_n > 0$ is the individual-specific marginal disutility of price (typically log-normally distributed to enforce the correct sign), $\mathbf{w}_n$ is the vector of individual WTP values drawn from a mixing distribution $f(\mathbf{w} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ with population mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and ε_{nti} is an IID Type I Extreme Value error. Because α_n and $\mathbf{w}_n$ are unobserved, the choice probability is obtained by integrating the logit kernel over the joint mixing distribution:

$$P_{nti}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \int\!\!\int \frac{\exp\!\left(-\alpha\left(p_{nti} - \mathbf{w}'\mathbf{x}_{nti}\right)\right)}{\sum_j \exp\!\left(-\alpha\left(p_{ntj} - \mathbf{w}'\mathbf{x}_{ntj}\right)\right)} \, f(\mathbf{w} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) \, g(\alpha) \, d\mathbf{w} \, d\alpha \tag{2}$$

This integral is evaluated by simulation using Halton draws (**trainHaltonSequencesNumerical1999**). Accounting for the panel structure of six repeated choice tasks per respondent, the simulated log-likelihood is:

$$\mathcal{L}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left[\frac{1}{R} \sum_{r=1}^{R} \prod_{t=1}^{6} \frac{\exp\!\left(-\alpha_r\left(p_{nti^*_{nt}} - \mathbf{w}_r'\mathbf{x}_{nti^*_{nt}}\right)\right)}{\sum_j \exp\!\left(-\alpha_r\left(p_{ntj} - \mathbf{w}_r'\mathbf{x}_{ntj}\right)\right)} \right] \tag{3}$$

where R is the number of Halton draws, $(\alpha_r, \mathbf{w}_r)$ is the r-th draw from the joint mixing distribution, and i^*_{nt} is the alternative chosen by respondent n in scenario t. Because estimation is carried out directly in WTP space, the population mean WTP for each non-price attribute k is obtained directly as $\hat{\mu}_k$ from the estimated distribution of $\mathbf{w}_n$, without requiring any post-estimation ratio transformation.

4.2. Latent Class Choice Model (LCCM)

Model Framework

We employ a LCCM to analyze consumer preferences and WTP for used BEVs. The LCCM is built on the premise that individuals can be grouped into a finite number of latent groups/segments/classes, within which individuals share homogeneous preferences, but across which preferences are heterogeneous (Greene and Hensher, 2003; Kamakura and Russell, 1989). This approach is particularly suited to our research context, where consumer valuations of battery health information likely differ systematically across distinct segments varied by individual characteristics.

As illustrated in Figure 3, the LCCM consists of two interrelated components which are estimated simultaneously. The *class membership model* assigns each respondent probabilistically to a latent class based on observable individual-level covariates and the *class-specific choice model* estimates the conditional probability of choosing each vehicle alternative within a given class. Additionally, a set of *inactive indicators*, including attitudinal and behavioral variables, though not directly included as estimators (due to collinearity, insignificance, or theoretical considerations), are used to further descriptively characterize the identified latent classes after estimation. The mathematical details of the LCCM are as follows, with notations consistent with Chen et al. (2023).

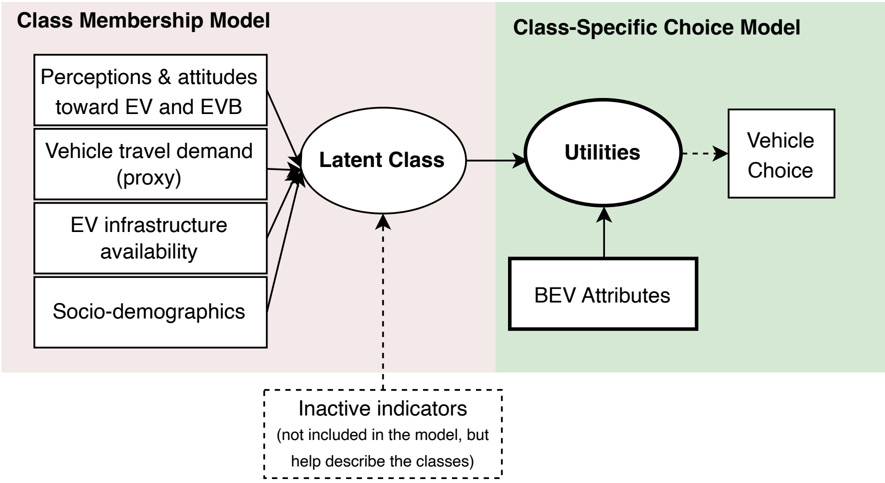

Figure 3: LCCM model framework.

Class-specific Choice Model

For respondent n belonging to latent class s ($s = 1, 2, \ldots, S$), the utility associated with choosing vehicle alternative i in choice scenario t ($t = 1, 2, \ldots, 6$)is specified as:

$$U_{nti|s} = \boldsymbol{\beta}_s' \mathbf{X}_{nti} + \varepsilon_{nti|s} \tag{4}$$

where $\mathbf{X}_{nti}$ is a vector of observed BEV attributes presented in scenario t to respondent n, including vehicle mileage, battery refurbishment status, estimated driving range and battery SOH currently (at years 3) and in the future (at year 8; 5 years from now), and purchase price; $\boldsymbol{\beta}_s$ is a vector of class-specific utility parameters capturing the marginal utility of each attribute for class s; and $\varepsilon_{nti|s}$ is a random error term assumed to be independently and identically distributed (IID) according to a Type I Extreme Value (Gumbel) distribution. Under this assumption, the conditional probability that respondent n in class s chooses alternative i in scenario t follows the multinomial logit (MNL) form:

$$P_{nt}(i \mid s) = \frac{\exp\left(\boldsymbol{\beta}_s' \mathbf{X}_{nti}\right)}{\sum_{j \in C_{nt}} \exp\left(\boldsymbol{\beta}_s' \mathbf{X}_{ntj}\right)} \tag{5}$$

where C_{nt} denotes the choice set available to respondent n at scenario t.

Class Membership Model

The class membership model specifies the probability that respondent n belongs to latent class s as a function of observed individual-level covariates. The class membership probability is:

$$M_n(s) = \frac{\exp\left(\boldsymbol{\gamma}_s' \mathbf{Z}_n\right)}{\sum_{s'=1}^{S} \exp\left(\boldsymbol{\gamma}_{s'}' \mathbf{Z}_n\right)} \tag{6}$$

where $\mathbf{Z}_n$ is a vector of individual-level covariates comprising four groups: (1) *perceptions and attitudes toward EVs and EV batteries* (e.g., range anxiety, environmental impact and functionality of EVB); (2) *vehicle travel demand proxies* (e.g., range of individual's primary vehicle); (3) *EV infrastructure availability* (e.g., access to electrical outlets); and (4) *socio-demographic characteristics* (e.g., income). $\boldsymbol{\gamma}_s$ is the vector of membership function parameters for class s. To ensure identification, the parameters for the reference class S are normalized to zero.

The unconditional probability that respondent n chooses alternative i in scenario t, integrated over all latent classes, is:

$$P_{nt}(y_n = i) = \sum_{s=1}^{S} P_{nt}(i \mid s) \cdot M_n(s) \tag{7}$$

Substituting Equations (5) and (6) into Equation (7), the full integrated choice probability is:

$$P_{nt}(y_n = i) = \sum_{s=1}^{S} \left[\frac{\exp\left(\boldsymbol{\beta}_s' \mathbf{X}_{nti}\right)}{\sum_{j \in C_{nt}} \exp\left(\boldsymbol{\beta}_s' \mathbf{X}_{ntj}\right)} \cdot \frac{\exp\left(\boldsymbol{\gamma}_s' \mathbf{Z}_n\right)}{\sum_{s'=1}^{S} \exp\left(\boldsymbol{\gamma}_{s'}' \mathbf{Z}_n\right)} \right] \tag{8}$$

The log-likelihood function across all N respondents and T choice scenarios is:

$$\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\gamma}) = \sum_{n=1}^{N} \sum_{t=1}^{T} \ln P_{nt}(y_n = i_n^*) \tag{9}$$

where i_n^* denotes the alternative chosen by respondent n in scenario t. The model parameters $\boldsymbol{\beta}_s$ and $\boldsymbol{\gamma}_s$ are estimated jointly via maximum likelihood estimation.

Willingness to Pay

A key output of the LCCM is class-specific WTP for each BEV attribute. Under the assumption that the marginal utility of price is negative and constant across alternatives, the WTP for attribute k in class s is:

$$\mathrm{WTP}_{s,k} = -\frac{\hat{\beta}_{s,k}}{\hat{\beta}_{s,\text{price}}} \tag{10}$$

This quantity represents the maximum dollar amount a representative respondent in class s would be willing to pay (or accept as compensation) for a one-unit improvement in attribute k, holding all other attributes constant.

Model Selection

The optimal number of latent classes is determined by estimating models with two through five classes and comparing fit using multiple information criteria: the log-likelihood (LL), Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and class size. Lower values of AIC, BIC, and CAIC indicate better model fit penalized for complexity. The preferred specification is selected based on the lowest BIC, acceptable entropy, and substantively interpretable class profiles of reasonable size.

5. Results and Discussion

5.1. Model Selection

5.2.

Class 1: Hostile Reluctant (13%)

Profile: Older, lower-income, ICEV-primary households with lower EV knowledge, short typical vehicle range, and low charging outlet access.

Mechanism of attribute insensitivity: High variance and noisy, inconsistent engagement. β_{price} is non-zero and significant enough to anchor WTP, but all attribute coefficients have large standard errors relative to point estimates. These respondents are not truly indifferent: they carry negative associations with BEVs (high range anxiety, low EV knowledge), but their choices are sufficiently inconsistent that the model cannot recover stable preference parameters. The near-zero opt-out WTP (−\$3.2k, ns) marks them as genuinely likely to walk away, unlike Class 4, who are also attribute-insensitive but strongly BEV-committed.

Class 2: Refurbishment-Averse (24%)

Profile: Higher income, EV-knowledgeable, environmentally positive, with charging outlet access. Fewer ICEV-primary households.

Mechanism: Genuine, strong preferences that are statistically recoverable. The defining characteristic is *negative* WTP for battery refurbishment: pack replacement costs this class \$14.9k in WTP, and cell replacement \$5.3k. This is counterintuitive, as refurbishment might be expected to signal a positive intervention. This class appears to interpret refurbishment as a signal of battery failure risk or future hassle cost rather than a benefit. They value range strongly and tolerate minor degradation (<12%) but are indifferent to severe degradation (24%+), possibly because they screen out vehicles with extreme degradation before purchase.

Comparison with Class 3: Both classes value range and oppose degradation, but differ critically on three dimensions: (1) Refurbishment: Class 2 penalizes, Class 3 values (pack replacement, +\$10.5k). (2) Severe degradation: Class 2 is indifferent (loss pw3, ns), Class 3 penalizes strongly (loss pw3, −\$6.2k, $p < 0.01$). (3) Opt-out aversion: Class 2 (−\$43k) vs. Class 3 (−\$16.6k), indicating Class 2 is substantially more committed to completing a BEV purchase.

Implication: Class 2 represents a sophisticated, high-income segment that views refurbishment as a reliability red flag. Messaging should emphasize preventive battery design and longevity guarantees rather than repair availability.

Class 3: Range-Focused Mainstream (42%)

Profile: Broad mainstream segment: middle income, moderate EV knowledge, mixed-fuel households, moderate outlet access. Largest class.

Mechanism: Stable, coherent, and economically interpretable preferences across all attributes. This is the prototypical BEV buyer who responds rationally to battery information. Range exhibits diminishing marginal value (pw2 > pw1 in WTP increment, but the slope flattens). Degradation carries a nonlinear penalty: mild loss is tolerated (pw1, \$12.2k), moderate loss less so (pw2, \$5.3k), and severe loss is actively penalized (pw3, −\$6.2k). Refurbishment is valued positively, consistent with viewing it as a consumer protection mechanism.

Comparison with Class 2: Class 3 penalizes severe degradation strongly; Class 2 does not. Class 3 values refurbishment; Class 2 penalizes it. Class 3 has lower opt-out aversion and is less wealthy and less EV-knowledgeable than Class 2.

Class 4: Budget-Constrained, Price-Sensitive

Profile: Lower income, lower outlet access, ICEV-primary households, shorter typical vehicle range, lower EV knowledge.

Mechanism, price-scaling suppression: $\beta_{\text{price}} = -4.27$ is approximately 4× larger in magnitude than Classes 2 and 3 ($\beta_{\text{price}} \approx -1.0$). Because WTP $= \beta_{\text{attr}}/(-\beta_{\text{price}})$, every attribute coefficient is divided by 4.27. A coefficient that would produce a \$10k WTP in Class 3 produces only \$2.3k here. The apparent attribute insensitivity in WTP terms is a mechanical artifact of extreme price sensitivity, not genuine indifference. The exception is loss pw3 (24%+ degradation), which breaks through because the underlying $\beta_{\text{loss,pw3}}$ is sufficiently large relative to the scaling, yielding a statistically significant penalty even after division. These respondents are genuinely sensitive to battery degradation; they simply cannot afford to pay much for any attribute.

Class 5: Attribute-Insensitive (17%)

Following the choice experiments, the survey includes a self-assessment question asking respondents to rate the importance of each attribute using a Likert scale, ranging from “Not at all important” to “Extremely important.” This question serves multiple purposes: (1) Validating choice behavior – It provides a way to cross-check whether the attributes respondents claim to value (i.e., stated importance) align with the trade-offs revealed in their actual choices (i.e., derived importance) (Chu, 2002; Huster et al., 2024). Inconsistencies may signal issues such as misunderstanding or inattention. (2) Understanding perceived salience – Because respondents may focus on different attributes during the experiment, self-reported ratings help identify which attributes were most influential in their decision-making process. (3) Segmenting respondents – These ratings can be used to group participants based on their stated priorities (e.g., price-sensitive vs. battery-conscious consumers), enabling more targeted analysis or class-specific modeling. (4) Survey diagnostics and data quality checks – Discrepancies between stated importance and revealed preferences may indicate low engagement, confusion, or misinterpretation of the attribute descriptions, helping researchers assess the overall quality of the data.

6. Conclusion

As more EVs reach the end of their first ownership cycle, understanding the dynamics of the used EV market, particularly the unique concerns surrounding battery condition, has become both timely and essential. Yet the lack of transparent, standardized battery health information at the point of sale continues to fuel uncertainty and undervaluation, which may impact adoption of pre-owned EVs. This DCI provides guidelines to survey design and data collection effort that could address a major gap in the literature by focusing on how consumers evaluate used battery electric vehicles (BEVs), what factors influence their willingness to adopt them, and how information asymmetries, particularly around battery health, impact market confidence and transaction outcomes.

This survey-based study represents a novel and rigorous approach to addressing these challenges. By combining insights from prior literature with a discrete choice experiment (DCE), the survey design allows researchers to empirically estimate the value consumers place on different forms of battery information and vehicle characteristics. Looking ahead, this DCI and the results from this research project can help shape the next phase of research and innovation in the second-life EV battery space. For example, further studies could explore consumer attitudes toward more complex refurbishment scenarios, such as partial vs. full battery replacements, and investigate how different warranty structures influence willingness to pay. Additionally, integrating insights from behavioral economics and data from telematics or vehicle diagnostics could enhance our understanding of how consumers interpret and act upon battery health information.

Latent class heterogeneity is driven by BEV aversion (the extent to which they reject BEVs at purchase) and disutility of battery degradation. Range anxiety are universal, including initial capacity and long-run loss. Battery cell-level and pack-level refurbishments are valued similarly despite their technical differences, implying limited consumer understanding.

CRediT authorship contribution statement

Xiatian Iogansen: Conceptualization, Data curation, Methodology, Software, Formal analysis, Writing - Original Draft, Writing - Review & Editing. **Zain hoda:** Conceptualization, Data curation, Methodology, Writing - Review & Editing. **Christina Gore:** Conceptualization, Data curation, Methodology, Investigation, Writing - Original draft preparation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Joshua D. Kneifel:** Conceptualization, Data curation, Methodology, Investigation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Sindhu Ranganath:** Data curation, Writing - Original Draft, Writing - Review & Editing. **John Helveston:** Conceptualization, Methodology, Writing - Review & Editing, Supervision.

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